

Problem-Solution Fit

Date	12 July 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural students • Farm advisors • Small & medium-scale agricultural businesses	6. CUSTOMER LIMITATIONS[CL] EG. BUDGET, DEVICES • Limited technical knowledge • Limited internet access • Budget constraints • Lack of awareness about AI tools	5. AVAILABLE SOLUTIONS[AS] PROS & CONS Pros: • Traditional farming knowledge • Agricultural experts • Government advisories Cons: • Time-consuming • Not personalized • May not use real-time data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty selecting the right crop. • Unpredictable weather affects yield. • Low crop productivity. • Financial losses due to wrong crop selection.	9. PROBLEM ROOT / CAUSE [RC] • Lack of accurate crop prediction. • Multiple environmental factors affect crop growth. • Limited access to data-driven recommendations.	7. BEHAVIOR + ITS INTENSITY [BE] • Depends on experience and local advice. • Checks weather forecasts. • Searches online for farming information. • Consults nearby farmers.
3. TRIGGERS TO ACT [TR] • Poor harvest in previous season. • Changes in weather conditions. • Need to improve crop yield. • Availability of AI-based farming tools.	10. YOUR SOLUTION [SL] • OptiCrop uses Machine Learning to analyze: • Nitrogen (N) • Phosphorus (P) • Potassium (K) • Temperature • Humidity • pH • Rainfall • It predicts the most suitable crop and provides confidence scores, helping farmers make informed decisions and improve crop productivity.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Website • Mobile browser • Agricultural websites
4. EMOTIONS[EM] BEFORE / AFTER Before: • Confused • Worried After: • Confident • Satisfied		OFFLINE • Farmers • Agricultural officers • Training programs